

Customer 360 - Approach Document

Volopay Growth Squad Assessment | July 2026

Customer 360 AI Assistant — Approach Document

Growth Squad Technical Assessment | Volopay | July 2026

1. Problem Statement

Customer Success teams at modern SaaS companies operate across multiple disconnected data systems — CRM platforms, support ticketing tools, email inboxes, and product analytics dashboards. The result is fragmented visibility: a CSM may know a customer's ARR but not their open critical ticket; a support agent may know about an escalation but not that renewal is 14 days away.

This fragmentation leads to reactive account management, missed upsell windows, and avoidable churn.

2. Business Challenge

Challenge	Impact
No unified account view	CSMs make decisions without full context
Manual health assessment	Inconsistent, subjective scoring
Reactive escalation	Critical issues surface too late
No next-action guidance	Valuable accounts receive generic treatment
LLM integration complexity	Teams unsure how to safely add AI to CS workflows

3. Solution

Customer 360 AI Assistant — a Streamlit application that aggregates four data sources into a single intelligent account view and applies a deterministic rule engine to generate:

- A weighted **Health Score** (0–100) with transparent breakdown
- A prose **Customer Summary** ready for internal briefing
- A prioritized **Business Risks** list
- An **Opportunities** list for upsell and expansion
- A single, context-aware **Next Best Action** with owner and urgency

The engine is fully functional without an API key, and has clearly marked # LLM_HOOK integration points for OpenAI GPT-4 or Google Gemini activation.

4. Architecture Diagram

```

flowchart TD
    subgraph DATA ["■ Data Sources"]
        A["crm.csv\n(CRM Records)"]
        B["support.csv\n(Support Tickets)"]
        C["emails.csv\n(Email History)"]
        D["usage.csv\n(Product Usage)"]
    end

    subgraph UTILS ["■ utils.py – Intelligence Engine"]
        E["load_all_data()\nData Loaders"]
        F["get_customer_profile()\nProfile Builder"]
        G["compute_health_score()\nWeighted Scoring Engine"]
        H["generate_customer_summary()\nProse Generator"]
        I["generate_business_risks()\nRisk Detector"]
        J["generate_business_opportunities()\nOpportunity Finder"]
        K["generate_next_best_action()\nPriority Action Engine"]
        L["LLM_HOOK: call_openai()\ncall_gemini()"]
    end

    subgraph APP ["■ app.py – Streamlit UI"]
        M["Sidebar\nCustomer Selector + AI Mode"]
        N["CRM Overview\nMetric Cards"]
        O["Support History\nStyled Ticket Table"]
        P["Email Summary\nBody + AI Summary"]
        Q["Usage Metrics\nProduct Engagement"]
        R["AI Insights Tabs\nSummary | Risks | Opps | NBA | Health"]
    end

    subgraph DEPLOY ["■ Deployment"]
        S["Streamlit Community Cloud"]
        T["Render Web Service"]
    end

    A --> E
    B --> E
    C --> E
    D --> E
    E --> F
    F --> G
    F --> H
    F --> I
    F --> J
    G & H & I & J --> K
    H & I & J & K -.->|Optional| L
    G & H & I & J & K --> APP
    M --> N & O & P & Q & R
    APP --> DEPLOY

```

5. Workflow

1. User selects an account from the sidebar dropdown
2. app.py retrieves cached DataFrames (load_all_data)
3. get_customer_profile() merges all four sources by customer_id
4. compute_health_score() applies weighted rule engine → score + breakdown
5. Four generators produce: Summary, Risks, Opportunities, Next Best Action
6. UI renders all sections with color-coded metrics and styled components
7. (Optional) If API key is set, LLM hooks replace deterministic generators

6. Technology Stack

Layer	Technology	Version
UI Framework	Streamlit	≥ 1.32.0
Data Processing	Pandas	≥ 2.1.0
Python	CPython	3.10+
Styling	Custom CSS via st.markdown	—
Charts	Streamlit native + openpyxl	—
LLM (optional)	OpenAI GPT-4o / Google Gemini 1.5 Pro	—
Deployment	Streamlit Community Cloud / Render	—

7. Prompt Used (LLM Integration)

When LLM mode is activated, the following prompt template is used:

You are a Customer Success Manager AI assistant. Based on the following customer profile data, write a concise 3-4 sentence executive summary that can be used for internal team briefings.

Customer: {company}
Plan: {plan} | ARR: \${arr:,}
Days to Renewal: {days_to_renewal}
Open Tickets: {open_tickets} | Critical: {critical_tickets}
Health Score: {health_score}/100 ({health_label})
Email Summary: {email_summary}
Feature Adoption Score: {feature_adoption_score}

Focus on:

- Current account status and risk level
- Most urgent action required
- Revenue implications

Keep the tone professional and data-driven.

8. Dummy Data Description

crm.csv (15 records)

Simulates a B2B SaaS CRM — company names drawn from realistic industries (logistics, fintech, pharma, manufacturing). Plans range from Starter (\$12K ARR) to Enterprise (\$300K ARR). Renewal dates are within the next 12 months to create urgency scenarios.

support.csv (20 tickets)

Covers all 15 customers with varying ticket counts (0–2 per customer). Includes Critical, High, Medium, and Low priorities. Sentiment labels (Positive/Neutral/Negative) simulate AI classification output.

emails.csv (15 records)

One email record per customer. Email bodies are realistic business communication including escalations, renewal inquiries, and positive feedback. Email summaries simulate an LLM-generated extraction.

usage.csv (15 records)

Monthly product engagement: invoices processed, payments made, active users, API calls, and a composite feature adoption score (0–100) that drives health scoring.

9. Sample Input

Customer Selected: Crimson Media Group (C009)

CRM: Enterprise | ARR: \$300,000 | Renewal: 31 Jan 2025 | Owner: Priya Sharma
Support: 2 tickets – both Critical, both Open, both Negative sentiment
Email: "We have two open critical issues and our team is running out of patience.
We process \$2M+ in payments monthly..."
Usage: 1,243 invoices | 23,450 API calls | Adoption: 95/100

10. Sample Output

Health Score: 20/100 — Critical

Customer Summary:

Crimson Media Group is on the Enterprise plan with an ARR of \$300,000, managed by Priya Sharma. Their renewal is in 42 days, making this an urgent account to prioritize. There are 2 critical support tickets currently open, which presents an escalation risk and requires immediate CSM attention. Overall account health is rated Critical (20/100).

Top Risk: ■ Critical Support Escalation — 2 critical tickets unresolved at \$300K ARR.

Next Best Action: Escalate 2 critical tickets to Engineering with VP-level visibility.

- Urgency: ■ Immediate (< 24 hrs)
- Owner: Priya Sharma

11. Future Improvements

Improvement	Description	Priority
LLM Integration	Full OpenAI / Gemini activation for narrative insights	High
Real CRM Sync	Connect to Salesforce / HubSpot via API	High
Predictive Churn Model	ML model trained on historical churn patterns	Medium
Email Ingestion	Auto-parse emails from Gmail / Outlook via OAuth	Medium
Slack Alerts	Push Next Best Action to CSM's Slack channel	Medium
Multi-language Support	Generate summaries in regional languages	Low

Mobile Optimization	Responsive layout for field CSM use	Low
A/B Testing	Test LLM vs rule-based output quality	Low

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